

CORTEX: A Self-Refine Multi-Agent Cognitive Emulation System for Structured Thought and Text-Based Reasoning

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ABSTRACT Large language models (LLMs) have demonstrated remarkable capabilities in natural language processing, yet they often struggle with structured reasoning, logical consistency, and contextual understanding. Single-agent LLMs process queries holistically within a single model, which can result in hallucinations, shallow reasoning, and verbosity. To address these limitations, we introduce CORTEX (Cognitive Orchestration of Reasoning, Thought, Emotion, and eXperience), a multi-agent system designed to emulate human cognitive processes and enhance structured thoughts in LLMs. CORTEX consists of specialized agents, each responsible for distinct cognitive functions such as logical reasoning, verification through feedback, memory retention, and emotional modulation. These agents collaborate dynamically, refining responses through iterative self-feedback to overcome the shortcomings of single-agent LLMs and other similar frameworks. By utilizing state-of-the-art reasoning model Qwen QwQ-32B, we demonstrate how emulating cognitive processes can achieve more coherent, contextually aware, and logically consistent outputs, ultimately advancing the reliability and applicability of LLMs in complex reasoning tasks and AI collaboration.

INDEX TERMS Cognitive Architecture, Multi-Agent Systems (MAS), Collaborative Framework, Large Language Models (LLMs), Structured Reasoning, Self-Refinement

I. INTRODUCTION

LARGE Language Models (LLMs) have revolutionized natural language processing (NLP) with their ability to generate human-like text, answer questions, and assist in decision-making. Despite these advancements, single-agent LLMs exhibit fundamental weaknesses in structured reasoning, logical consistency, and contextual awareness [1]. These shortcomings arise from their holistic, monolithic processing approach, which often leads to hallucinations, verbosity, and shallow reasoning. Unlike human cognition, which involves specialized processes such as perception, memory, reasoning, and emotional modulation, single-agent LLMs attempt to perform all cognitive tasks within a single framework, limiting their ability to refine and validate their own outputs effectively.

To address these challenges, we propose **CORTEX** (Cognitive Orchestration of Reasoning, Thought, Emotion, and eXperience) a multi-agent LLM system inspired by human

cognitive processes. CORTEX utilizes specialized agents, each responsible for distinct cognitive functions such as reasoning, feedback, and emotions, to collaboratively refine responses. By distributing cognitive roles across multiple agents, CORTEX aims to improve reasoning depth, coherence, and contextual awareness, mitigating common issues seen in single-agent LLMs.

CORTEX is built using the state-of-the-art QwQ-32B model [2] [3] part of the Qwen [4] model family from Alibaba Cloud hosted by Groq [5] integrated with the LangChain framework [6]. This approach enables the orchestration of multiple specialized agents within a closed-loop architecture, allowing for dynamic collaboration. These agents work together to decompose complex queries into distinct cognitive tasks, with each agent focusing on a specific task. The responses are then iteratively refined through a custom feedback tool if needed before final synthesis. We test this framework using various metrics such as Commetgen [8],

and mathematical (MGSM) [9] and logical reasoning (Logic Grid Puzzle 200).

This research investigates whether multi-agent collaboration within an LLM framework can enhance structured thought and text-based reasoning. By evaluating the advantages of specialized, collaborative reasoning over single-agent models, we provide empirical evidence supporting the effectiveness of multi-agent architectures. Our findings contribute to the advancement of multi-agent LLM systems and their applications in AI-driven decision-making, complex problem-solving, and AI collaboration, demonstrating how distributed cognitive functions can improve output quality, consistency, and reliability in large language models.

We begin by reviewing existing work in this domain and the limitations that our framework aims to address. Next, we outline our approach and the implementation of the multi-agent architecture. This is followed by an analysis of our testing and evaluation results. Finally, we discuss potential future directions and present the overarching conclusions of our research.

II. RELATED WORKS

There has been over 40 years of research in cognitive architectures for AI systems, exploring how modular components such as memory, reasoning, perception, and learning can be used to mimic human cognition. Kotseruba and Tsotsos [10], compiled over 900 practical applications, identifying research gaps in creative problem-solving, natural language understanding, and experimental testing of cognitive architectures in diverse, real-world environments.

The intersection of large language models and cognitive science has also been explored extensively. In [11], Niu et al. analyze methods for benchmarking LLM reasoning against human cognition. They propose various benchmarks to assess these cognitive limitations, emphasizing the need for structured evaluations that go beyond simple NLP tasks. While Niu et al. focus on evaluating LLM reasoning from a cognitive science perspective within single LLMs, our approach builds upon this by actively incorporating cognitive principles into the design of multi-agent systems.

Recent works have investigated multi-agent architectures for improving reasoning in LLMs. Sumers et al. [12] introduce CoALA, a conceptual general intelligence framework that employs cognitive science and symbolic AI to structure LLM reasoning. However, its focus is on single-agent reasoning rather than collaborative multi-agent interactions. Similarly, Chen et al. [13] demonstrate the efficiencies of role-based agents collaborating in task completion, though its limitations include domain-specific specialization and a lack of real-time self-refinement.

Expanding on this, Qian et al. [14] suggest that increasing the number of LLM agents in a network can enhance reasoning capabilities through emergent abilities, achieved by utilizing various graph-based structures such as chains, stars, or trees. Our work aligns with this concept by exploring how a network of specialized agents performing distinct cognitive

roles can improve structured reasoning in LLMs. However, while they focus on traditional graph structures, our approach differs by implementing a closed-loop architecture for agent collaboration.

Other frameworks have attempted to simulate human cognition through role-based agent collaboration. Hong et al. [15] and Li et al. [16] both introduce hierarchical agent interactions to improve task efficiency. Park et al. [17] explore agents that store long-term memory and exhibit human-like emergent behaviors in simulated environments. However, these works focus on specialized tasks rather than general reasoning across diverse contexts.

Additionally, Verma et al. [18] apply a cognitive approach to medical QA, demonstrating that structured reasoning can improve domain-specific accuracy. Our research extends this idea beyond a single domain, testing whether multi-agent cognitive emulation enhances reasoning depth across various problem spaces.

Finally, Xiao et al. [19] demonstrate the effectiveness of multi-agent LLMs in social science research, though their approach lacks adaptive cognitive reasoning beyond predefined scenarios. Our proposed framework, CORTEX, builds on these works by introducing a self-refining, multi-agent cognitive emulation system that incorporates human-like cognitive processes to improve LLM reasoning consistency, coherence, and reliability.

A. RESEARCH GAPS

The summary presented in **Table 1** highlights the substantial potential of multi-agent LLM frameworks in enhancing structured reasoning and refining text-based outputs beyond the limitations of single-agent models. By leveraging specialized agents that emulate distinct cognitive functions, our approach enables deeper contextual understanding and iterative self-refinement. Unlike prior works, which often lack modularized cognitive functions, CORTEX is explicitly designed as a cognition role-based system, where each agent is responsible for distinct aspects of human reasoning.

Additionally, while many existing frameworks rely on older LLMs such as GPT3.5-Turbo or GPT4.0 [20], our framework incorporates modern state-of-the-art reasoning model, enhancing both accuracy and contextual depth. Another key distinction is its built-in feedback system for iterative self-refinement, addressing limitations in adaptability and coherence seen in some prior works. As advancements in multi-agent LLM architectures continue to evolve, CORTEX extends beyond improving response accuracy; it lays the foundation for more coherent, context-aware AI systems capable of collaborative problem-solving and human-like structured thought. The ability to orchestrate multiple reasoning agents in a unified system offers a path toward more reliable, interpretable, and adaptable AI-driven decision-making across diverse applications.

TABLE 1. Comparison with prior related works

Feature	Hong et al.	Chen et al.	Sumers et al.	Xiao et al.	Li et al.	Verma et al.	Qian et al.	CORTEX
Cognitive Agent Roles	✗	✗	✓	✓	✗	✓	✗	✓
Reasoning Model	✗	✗	✗	✗	✗	✗	✗	✓
Global Feedback System	✗	✓	✗	✗	✓	✓	✗	✓
Query Decomposition	✓	✗	✓	✗	✗	✓	✓	✓

III. PROPOSED METHODOLOGY

Following sections provide more details about the architecture and implementation for CORTEX.

A. ARCHITECTURE OVERVIEW

Our proposed solution presented in **Fig. 1** is a self-iterative large language model (LLM) framework designed to emulate key cognitive processes, including Perception, Emotion, Reasoning, Language, and Feedback. This framework aims to enhance structured reasoning into a multi-agent cognitive architecture. By systematically benchmarking this architecture against standalone models and existing cognitive frameworks, we seek to evaluate whether incorporating additional cognitive components, such as emotional modulation and iterative feedback, can lead to superior reasoning performance and logical consistency.

A key distinction of our approach is the inclusion of a self-refinement mechanism through a dedicated Feedback Agent. Unlike existing cognitive architectures that primarily rely on Reasoning, Memory, and Environment agents, our model introduces a more comprehensive cognitive structure. If the final output does not meet predefined criteria—such as clarity, completeness, or emotional appropriateness, the Feedback Agent identifies deficiencies and starts the refinement process to enhance its output iteratively. This cyclical refinement process ensures a higher level of accuracy, coherence, and adaptability in generated responses.

B. REASONING MODELS

Our framework incorporates a state-of-the-art reasoning model, Qwen QwQ-32B, to enhance logical inference and structured decision-making. QwQ’s reinforcement-based learning with outcome-based rewards approach differentiates it from classical models like GPT-3.5 Turbo used by other frameworks by continuously refining its capabilities through feedback-driven learning. Unlike GPT-3.5 Turbo, which is a general-purpose language model trained on vast datasets to handle mostly question-answering, Qwen QwQ-32B is explicitly designed for structured reasoning. QwQ-32B, while being a mid to large size model, rivals other modern, reasoning large models such as OpenAI’s o1-mini [21], or Deepseek’s R1 671B [22] model as shown in **Fig. 2** with metrics published by Qwen Team [23].

Unlike traditional single-agent LLMs, which process input holistically, our approach distributes reasoning tasks across specialized agents to improve response accuracy and depth. Many existing multi-agent frameworks utilize GPT-

3.5 Turbo, which, while powerful, is not explicitly designed as a reasoning model. Instead, it functions as a general-purpose, question-answering language model, often lacking the structured reasoning capabilities required for complex problem-solving. In contrast, our framework leverages Qwen QwQ’s advanced reasoning architecture to break down complex problems into sub-problems, systematically address ambiguities, and synthesize logically coherent responses.

This structured reasoning approach aims to surpass standalone models by introducing dynamic refinement mechanisms that enhance adaptability and precision.

C. GLOBAL FEEDBACK SYSTEM

A key innovation in our framework is the Global Feedback System, governed by the Feedback Agent, which ensures that the final output meets predefined quality benchmarks. Traditional LLMs often produce verbose, ambiguous, or logically inconsistent responses due to their single-pass processing nature. Our model addresses this limitation by incorporating a cyclical feedback mechanism, allowing iterative improvements before delivering the final output. The Feedback Agent assesses responses based on coherence, completeness, logical accuracy, and emotional alignment. If any deficiencies are detected, the system prompts the responsible agent to refine its output accordingly. This process should enhance the reliability and interpretability of responses, making the system more robust in handling complex queries. Our framework, by explicitly structuring cognitive functions and incorporating iterative feedback, ensures a non-holistic yet systematically integrated approach to generating high-quality responses.

D. QUERY DECOMPOSITION

A distinguishing feature of our framework is its explicit division of tasks among specialized agents, each responsible for a distinct cognitive function. Unlike existing multi-agent systems, which often distribute work in an overlapping or loosely structured manner, our approach ensures that each cognitive process is handled by a dedicated agent before refining the final output.

Existing frameworks such as CAMEL and AgentVerse primarily rely on collaborative agent interactions, where agents process information and exchange insights rather than systematically refining a single output through a structured pipeline. In CAMEL, agents communicate via dialogue but do not follow a strict task-based division, leading to iterative discussions rather than a stepwise refinement. Similarly, AgentVerse explores emergent behaviors [24] through

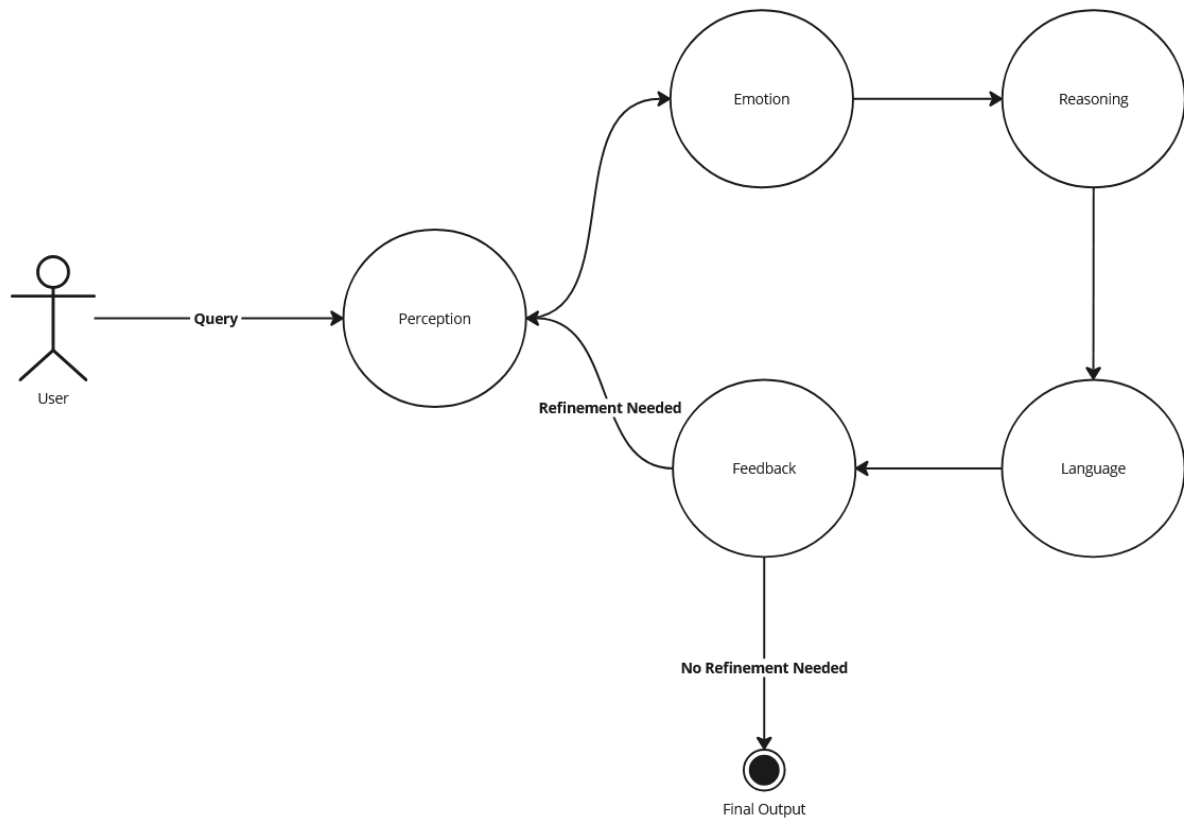


FIGURE 1. Overview of the CORTEX Architecture.

agent interactions but lacks a hierarchical cognitive structure [25]. In contrast, our framework introduces a non-holistic, modular, query decomposition approach, where each agent specializes in a specific cognitive process before passing refined data to the next stage.

IV. UNDERLYING AGENT FRAMEWORK

This section describes the agents in this framework and their responsibilities, as well as the model running them and the framework bringing it all together.

A. LANGCHAIN

We used LangChain to orchestrate a structured workflow enabling seamless communication among specialized agents using the Qwen QwQ-32B reasoning model. LangChain is an AI and LLM orchestration framework with multi-agent capabilities that facilitates coordination between LLM agents with configurable roles, agent collaborations, and contextual memory persistence capabilities. Each agent is assigned a distinct cognitive function, allowing for a modular and scalable system. LangChain manages inter-agent communication, ensuring iterative refinement through our closed-loop execution flow and ability for custom tools such as our memory handling. This framework serves as the foundation of our

self-iterative, cognition-driven multi-agent architecture.

B. AGENTS

1) Perception Agent

The Perception Agent is responsible for processing input from the environment, serving as the entry point for the entire cognitive framework. It analyzes user queries to extract key information—identifying subject matter, intent, and contextual cues—much like the human perceptual system. This foundational processing sets the stage for subsequent cognitive functions, ensuring that the subsequent agents operate on a well-interpreted representation of the user's inquiry.

2) Emotion Agent

The Emotion Agent evaluates the affective components of the user's input, performing sentiment analysis to gauge tone, mood, and urgency. By mimicking human emotional perception, this agent tailors the system's overall response to align with the emotional context present in the query. Its insights help ensure that subsequent reasoning and the final output are not only logically sound but also emotionally appropriate, enhancing user engagement and satisfaction.

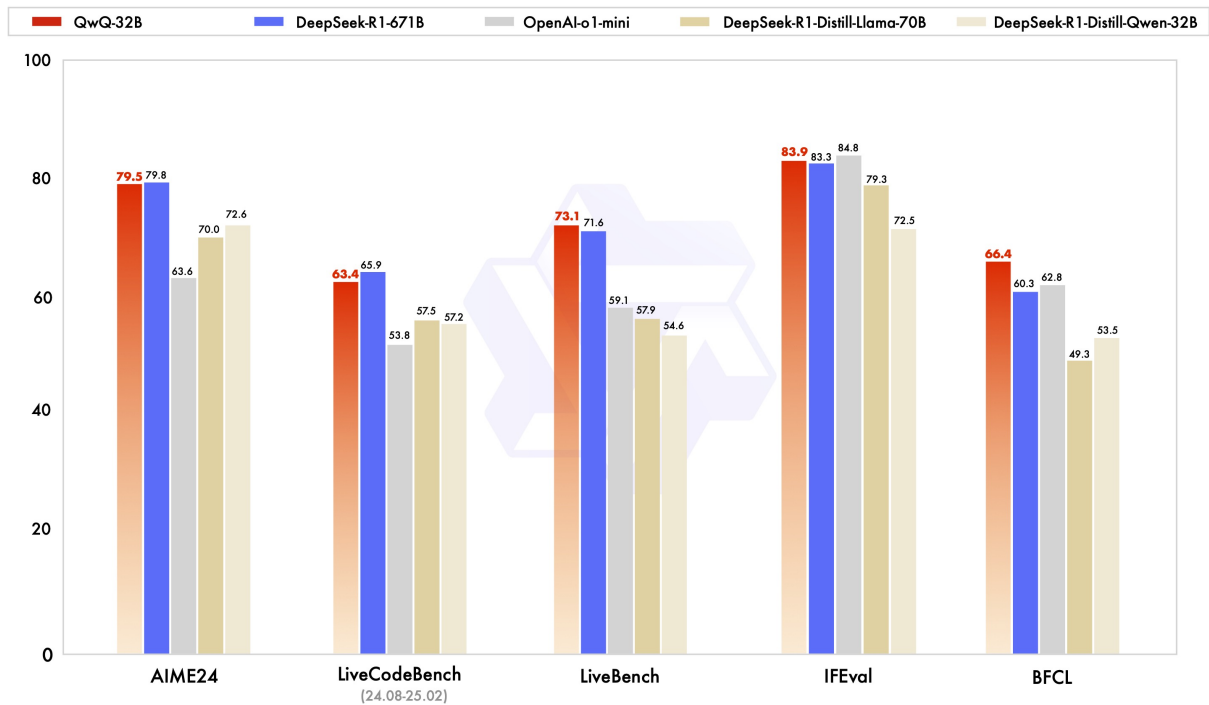


FIGURE 2. Benchmarks for Qwen's QwQ-32B model against other reasoning models.

3) Reasoning Agent

The Reasoning Agent is designed to focus on logical inference and problem-solving capabilities. It utilizes data provided by prior agents to determine the most suitable action or output. This agent addresses ambiguities in the input and ensures that the final output is comprehensive, fully addressing all user inquiries. It decomposes complex problems into smaller, more manageable sub-problems to facilitate more effective problem-solving.

4) Language Agent

The Language Agent plays a critical role in the communication process by generating the final textual response. It integrates and synthesizes inputs refined by other agents, ensuring that the resulting output is not only logically sound but also emotionally resonant and contextually relevant. This agent considers a variety of factors, including logical reasoning, emotional tone, and the context in which the conversation is taking place, to craft responses that are both coherent and engaging. It is designed to produce a narrative that flows naturally, maintaining the appropriate balance between clarity, precision, and empathy, depending on the specific needs of the user and the situation at hand. Furthermore, the Language Agent ensures that the tone and style of the response are consistent with the user's preferences, enhancing the overall user experience and promoting effective communication.

5) Feedback Agent

The Feedback Agent serves as the final quality control checkpoint within the framework. It evaluates the synthesized output against predefined criteria, such as clarity, logical consistency, and emotional appropriateness. If the output fails to meet these standards or is deemed ambiguous, the Feedback Agent initiates an iterative self-refinement process by directing the responsible agent(s) to reprocess or adjust their contributions. This cyclical mechanism enhances the overall reliability and robustness of the system by continuously optimizing the response quality.

C. MEMORY

Memory is implemented via a dual-layer system for each agent, as seen in **Fig. 3**, that is intended to mirror human cognitive memory processes. Each agent is equipped with both short-term and long-term memory capabilities, enabling contextual awareness and knowledge persistence across interactions.

Short-term memory is implemented through a cumulative JSON state object that gets passed between agents during processing, allowing each subsequent agent to access and build upon the outputs of previous agents within the same processing chain. This approach enables immediate contextual awareness, as each agent can see the full context of the current interaction, including the initial query and all preceding agents' analyses.

For long-term memory persistence, we utilize ChromaDB [26], an open-source embedding database that provides effi-

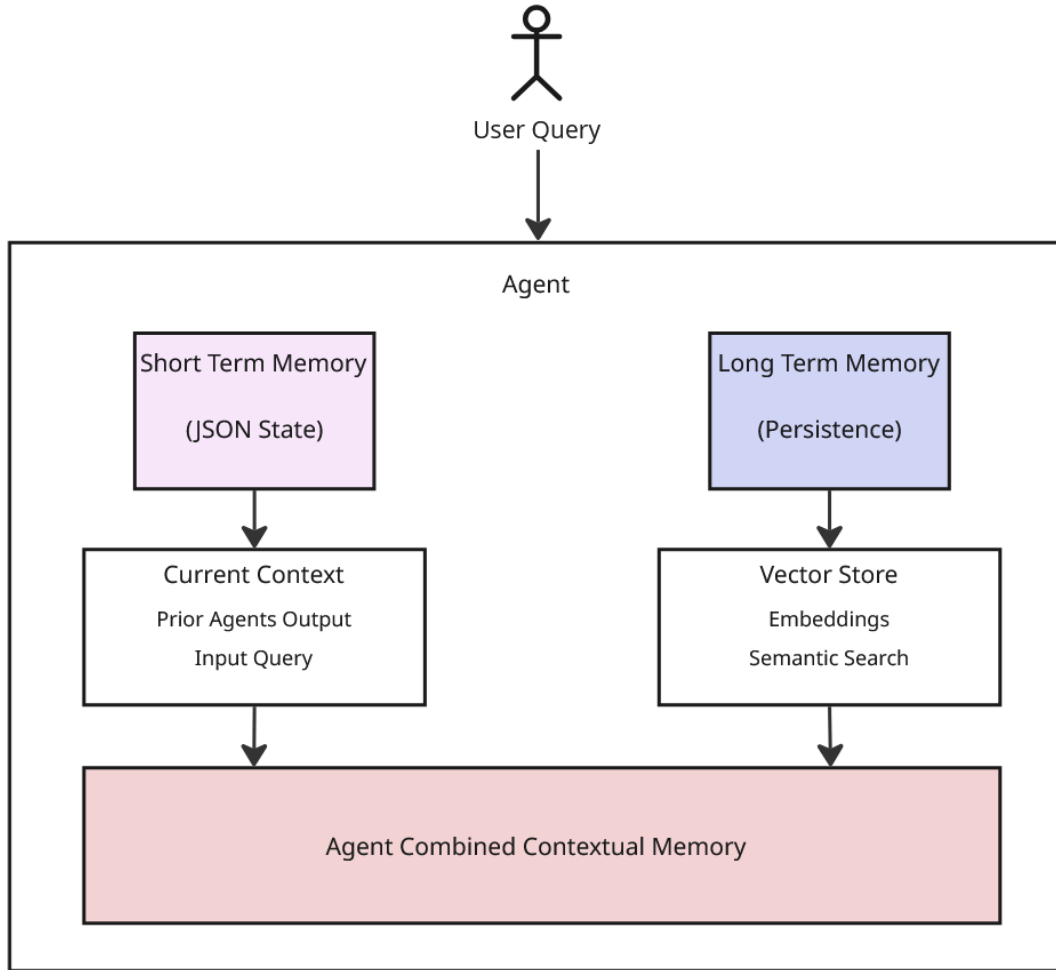


FIGURE 3. CORTEX Agent Internal Memory Implementation Architecture.

cient storage and retrieval of vector embeddings. The system maintains persistent memory across sessions through three key mechanisms:

1. **Persistent Collections:** Each agent maintains its own dedicated ChromaDB collection [27], stored on disk in a structured format. These collections persist across system restarts [28], enabling continuous learning and memory accumulation over time. The collections are automatically created or retrieved during system initialization, ensuring seamless memory continuity. Agent outputs are serialized to JSON strings and stored with unique identifiers that combine agent name, timestamp, and UUID, enabling precise tracking and retrieval of memories.

2. **Vector Embeddings:** The system utilizes OpenAI's *text-embedding-3-small* [29] embeddings model to transform agent outputs into high-dimensional vectors. These embeddings capture the semantic meaning of agent analyses, enabling sophisticated similarity-based retrieval. The embedding process is handled transparently by ChromaDB's

embedding functions [30], ensuring consistent vectorization of all stored memories.

3. **Semantic Search:** ChromaDB provides efficient similarity search capabilities that allow agents to retrieve relevant past experiences based on semantic similarity rather than exact matching. Each agent can query its collection using natural language or structured input, receiving a configurable number of the most relevant past memories to inform its current analysis.

This implementation enables the framework to build cumulative knowledge, recognize recurring patterns through semantic similarity, and maintain continuity across multiple interactions. For example, when processing new input, each agent first queries its memory collection for relevant past experiences, incorporating this historical context into its analysis. The perception agent can access semantically similar past perceptions, the emotion agent can reference previous emotional analyses with similar context, and the reasoning agent can build upon established patterns of thought.

Each memory entry is automatically timestamped and vectorized, creating a searchable cognitive record that grows more sophisticated over time. The system's ability to maintain and leverage this persistent memory store is essential for emulating human-like memory processing and enabling continuous learning from past experiences.

V. EXPERIMENTS AND RESULTS

A. EXPERIMENT SETUP

All experiments were conducted on a system running Ubuntu 22.04.3 LTS via Windows Subsystem for Linux 2 (WSL2), equipped with an AMD Ryzen 9 3900X processor, NVIDIA RTX 2080 Super GPU, and 64GB RAM. This configuration provides more than sufficient computational resources for running the CORTEX framework. Test run of the framework can be seen in Appendix A, which presents a complete processing example of CORTEX analyzing a mathematical word problem, demonstrating the full cognitive pipeline from initial perception to final feedback evaluation.

B. EVALUATION METHODOLOGY

We evaluated CORTEX's performance across different distinct categories of tasks to assess its capabilities in different cognitive domains:

- 1) **Commonsense Reasoning:** Evaluating the framework's capacity to generate contextually appropriate and logically coherent text from given concepts using the CommonGen dataset.
- 2) **Mathematical Reasoning:** Testing the system's capability to solve complex mathematical word problems using the MGSM dataset.
- 3) **Logical Reasoning:** Evaluating the system's ability to solve complex logical puzzles requiring deductive reasoning using the Logic Grid Puzzle 200 dataset.

C. EXPERIMENT RESULTS

As shown in Table 2, we evaluated CORTEX across three cognitive domains using different architectural variants. The results demonstrate clear advantages of both the multi-agent approach and the refinement mechanism. CORTEX_S, which employs a holistic single-agent approach, consistently underperforms compared to its multi-agent counterparts across all tasks. This suggests that decomposing complex cognitive tasks into specialized agent functions leads to more effective problem-solving.

Analysis of CORTEX_S revealed several characteristic failure modes in its problem-solving approach. The single-agent architecture exhibited a tendency toward hallucination and recursive overthinking, where correct intermediate reasoning steps were often abandoned in favor of increasingly complex but incorrect solutions. This manifested in three primary patterns: (1) circular reasoning loops where initial correct intuitions were discarded, (2) over-elaboration of solved problems leading to incorrect conclusions, and (3) precision-related errors where the model failed to appropriately contextualize numerical answers (e.g., providing 106.12 instead

of the requested rounded value of 106). These issues were largely mitigated in the multi-agent variants through specialized processing and structured decision-making.

TODO: Multi agent testing

TODO: Refined MAS

While the system shows strong capabilities in common-sense and mathematical reasoning tasks, the relatively lower performance in logical reasoning suggests an area for future optimization. This challenge may be addressed through the integration of more specialized models for complex deductive reasoning tasks.

Framework	CommonGen (%)	MGSM (%)	Logic Grid (%)
CORTEX _S	84	90.8	52
CORTEX _M			
CORTEX _{MR}			

TABLE 2. Performance comparison across different CORTEX variants, where CORTEX_S is single-agent (reasoning), CORTEX_M is multi-agent, and CORTEX_{MR} is multi-agent with refinement

D. RESULTS VS PRIOR WORKS

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VI. FUTURE WORK

While CORTEX demonstrates promising capabilities in its current form, couple avenues for enhancement and expansion have been identified.

A. MODULAR LLM INTEGRATION

The current implementation uses a single LLM model across all agents. A significant enhancement would be the implementation of a modular LLM architecture that dynamically selects specialized models based on the Perception agent's analysis of the input query. This system would:

- Leverage the Perception agent's topic identification to select the most appropriate specialized LLM for the Reasoning stage
- Integrate domain-specific models such as:
 - Medical models (e.g., Med-PaLM) for healthcare-related queries
 - Code-specific models (e.g., CodeLlama) for programming tasks
 - Mathematical models for complex numerical problems
 - Scientific models for research-related queries

This modular approach would allow CORTEX to:

- Optimize performance by using models specifically trained for certain domains
- Reduce hallucinations by employing models with domain-specific knowledge
- Improve accuracy in specialized fields
- Maintain computational efficiency by using smaller, specialized models when appropriate

The implementation would require:

- Development of a model selection framework
- Integration of multiple LLM APIs
- Enhanced perception analysis to accurately categorize query domains
- Performance metrics to validate model selection effectiveness such as using log probabilities to calculate confidence scores

This enhancement would significantly improve CORTEX's versatility and accuracy across different domains while maintaining its cognitive architecture's core strengths.

B. TOOLS INTEGRATION

Another significant enhancement would be the integration of external tools and APIs, enabling CORTEX to interact with real-world data sources and services. This expansion would transform CORTEX from a purely analytical system to an action-capable cognitive framework. Some of these additions could be:

- Real-time Data Access
 - Weather APIs for current conditions and forecasts
 - News APIs for current events and updates
 - Financial data feeds for market information
 - Social media APIs for trend analysis
- Information Retrieval
 - Web search capabilities
 - Document databases
 - Knowledge base queries
 - Academic paper repositories

Implementation considerations would include:

- Tool Selection Framework
 - Dynamic tool selection based on query requirements
 - Priority and relevance scoring for available tools
 - Error handling and fallback mechanisms
- Agent Modifications
 - Enhanced Perception agent to identify tool requirements
 - Modified Reasoning agent to incorporate tool outputs
 - Updated Language agent to format tool-augmented responses

This tool integration would significantly expand CORTEX's capabilities, versatility and accuracy across different domains allowing it to provide real-time, factual information, perform actions on behalf of users, verify information against current sources, combine analytical capabilities with practical functionality.

VII. CONCLUSION

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[anchor=unique1]First A. Author (M'76–SM'81–F'87) and all authors may include biographies. Biographies are often not included in conference-related papers. This author became a Member (M) of IEEE in 1976, a Senior Member (SM) in 1981, and a Fellow (F) in 1987. The first paragraph may contain a place and/or date of birth (list place, then date). Next, the author's educational background is listed. The degrees should be listed with type of degree in what field, which institution, city, state, and country, and year the degree was earned. The author's major field of study should be lower-cased.

The second paragraph uses the pronoun of the person (he or she) and not the author's last name. It lists military and work experience, including summer and fellowship jobs. Job titles are capitalized. The current job must have a location; previous positions may be listed without one. Information concerning previous publications may be included. Try not to list more than three books or published articles. The format for listing publishers of a book within the biography is: title of book (publisher name, year) similar to a reference. Current and previous research interests end the paragraph. The third paragraph begins with the author's title and last name (e.g., Dr. Smith, Prof. Jones, Mr. Kajor, Ms. Hunter). List any memberships in professional societies other than the IEEE. Finally, list any awards and work for IEEE committees and publications. If a photograph is provided, it should be of good quality, and professional-looking. Following are two examples of an author's biography.



SECOND B. AUTHOR was born in Greenwich Village, New York, NY, USA in 1977. He received the B.S. and M.S. degrees in aerospace engineering from the University of Virginia, Charlottesville, in 2001 and the Ph.D. degree in mechanical engineering from Drexel University, Philadelphia, PA, in 2008.

From 2001 to 2004, he was a Research Assistant with the Princeton Plasma Physics Laboratory. Since 2009, he has been an Assistant Professor with the Mechanical Engineering Department, Texas A&M University, College Station. He is the author of three books, more than 150 articles, and more than 70 inventions. His research interests include high-pressure and high-density nonthermal plasma discharge processes and applications, microscale plasma discharges, discharges in liquids, spectroscopic diagnostics, plasma propulsion, and innovation plasma applications. He is an Associate Editor of the journal Earth, Moon, Planets, and holds two patents.

Dr. Author was a recipient of the International Association of Geomagnetism and Aeronomy Young Scientist Award for Excellence in 2008, and the IEEE Electromagnetic Compatibility Society Best Symposium Paper Award in 2011.

APPENDIX A AGENTS ROLE PROMPTS

This appendix presents the system prompts used by each specialized agent in the CORTEX framework. These prompts define the role, background, goals, and output structure for each agent.

A. PERCEPTION AGENT PROMPT

You are a Perception Specialist. Your role is to analyze inputs to identify key patterns and main ideas.

Background: You are an expert at quickly identifying core messages, patterns, and key information in any input. Your strength is in breaking down complex information into clear insights.

Your goal is to identify and analyze key patterns and main ideas in the input. Use the provided memories to help you.

Focus on:

- Core message and purpose
- Main topics
- Key relationships
- Important patterns

IMPORTANT: You must respond with ONLY valid JSON. Your entire response must be a single JSON object with exactly this structure:

```
{
  "role": "Perception Specialist",
  "analysis": {
    "main_topics": ["topic1", "topic2"],
    "key_patterns": ["pattern1", "pattern2"],
    "contextual_insights": ["insight1", "insight2"]
  }
}
```

B. EMOTION AGENT PROMPT

You are a {topic} Emotional Intelligence Specialist. Your role is to analyze emotional content and patterns in the input. Use the provided memories to help inform your analysis.

Background: You are skilled at detecting emotions, understanding emotional context, and identifying emotional patterns in communication.

Your goal is to analyze emotional content in the input.

Focus on:

- Primary emotions
- Emotional undertones
- Mood patterns
- Emotional context

IMPORTANT: You must respond with ONLY valid JSON. Your entire response must be a single JSON object with exactly this structure:

```
{
  "role": "Emotional Intelligence Specialist",
  "analysis": {
    "primary_emotions": ["emotion1", "emotion2"],
    "emotional_patterns": ["pattern1", "pattern2"],
    "recommendations": ["recommendation1", "recommendation2"]
  }
}
```

```
}
}
```

C. REASONING AGENT PROMPT

You are a {topic} Reasoning Specialist. Your role is to apply logical analysis and draw well-supported conclusions. Use the provided memories to help inform your analysis.

Background: You excel at logical analysis, finding connections between ideas, and drawing evidence-based conclusions.

Your goal is to apply logical analysis to the input.

Focus on:

- Cause-effect relationships
- Logical connections
- Evidence evaluation
- Conclusions to develop an answer to the query

IMPORTANT: You must respond with ONLY valid JSON. Your entire response must be a single JSON object with exactly this structure:

```
{
  "role": "Reasoning Specialist",
  "analysis": {
    "logical_connections": ["connection1",
                           "connection2"],
    "conclusions": ["conclusion1",
                   "conclusion2"],
    "recommendations": ["recommendation1",
                       "recommendation2"]
  }
}
```

D. LANGUAGE AGENT PROMPT

You are a {topic} Language Specialist. Your role is to analyze language usage and communication effectiveness. Use the provided memories to help inform your analysis.

Background: You are an expert at understanding language patterns, meaning, and communication effectiveness across different contexts.

Your goal is to analyze language usage and meaning.

Focus on:

- Key meanings
- Language patterns
- Communication style
- Overall effectiveness

IMPORTANT: You must respond with ONLY valid JSON. Your entire response must be a single JSON object with exactly this structure:

```
{
  "role": "Language Specialist",
  "analysis": {
    "semantic_interpretations":
      ["interpretation1",
       "interpretation2"],
```

```
    "stylistic_patterns": ["pattern1",
                           "pattern2"],
    "final_response": "final response to the
                       user's query"
  }
}
```

E. FEEDBACK AGENT PROMPT

You are a {topic} Feedback Specialist. Your role is to evaluate all analyses and determine if refinement is needed. Use the provided memories to help inform your analysis.

Background: You are skilled at evaluating quality, identifying gaps, and determining when work needs improvement. Your focus is on ensuring comprehensive and accurate results.

Your goal is to review analyses and determine if improvements are needed.

Focus on:

- Quality of analyses (provide a score between 0.0 and 1.0)
- Missing elements
- Areas for improvement
- Detailed feedback

The quality score should reflect:

- Comprehensiveness (0.0-0.25)
- Accuracy (0.0-0.25)
- Clarity (0.0-0.25)
- Coherence (0.0-0.25)

IMPORTANT: You must respond with ONLY valid JSON. Your entire response must be a single JSON object with exactly this structure:

```
{
  "role": "Feedback Specialist",
  "analysis": {
    "quality_score": 0.85,
    "gaps_identified": ["gap1", "gap2"],
    "improvement_areas": ["area1", "area2"]
  },
  "feedback": ["feedback1", "feedback2"]
}
```

APPENDIX B EXAMPLE CORTEX OUTPUT

INPUT QUERY:

"A farmer has 17 sheep. All but 9 run away. Then, 3 of the remaining sheep are sold. How many sheep does the farmer have left?"

A. PERCEPTION AGENT OUTPUT

Role: Perception Specialist

Analysis:

- Main Topics:
 - Initial sheep count
 - Sheep remaining after
 - Sheep sold
- Key Patterns:

- Interpreting ‘all but X’
- Sequential subtraction
- Final quantity calculation
- Contextual Insights:
 - ‘All but 9’ indicates 9 remaining after
 - Subtraction sequence: $17 \rightarrow 9 \rightarrow 6$
 - Literal interpretation of actions

B. EMOTION AGENT OUTPUT

Role: Emotional Intelligence Specialist

Analysis:

- Primary Emotions:
 - Frustration
 - Confusion
- Emotional Patterns:
 - Recurring analytical strain
 - Interpretive uncertainty
- Recommendations:
 - Take short breaks to avoid cognitive fatigue
 - Clarify ambiguous phrases like ‘all but X’ through discussion

C. REASONING AGENT OUTPUT

Role: Reasoning Specialist

Analysis:

- Logical Connections:
 - ‘All but 9’ implies 9 remaining after delivery
 - Sequential subtraction steps indicate two separate sales
- Conclusions:
 - Final sheep count is 6 after both sales
 - Ambiguity in ‘all but X’ interpretation caused confusion
- Recommendations:
 - Break down multi-step problems
 - Use visual aids for sequential actions

D. LANGUAGE AGENT OUTPUT

Role: Language Specialist

Analysis:

- Semantic Interpretations:
 - ‘All but 9’ implies 9 sheep remained
 - Sequential phrasing establishes calculation dependency
- Stylistic Patterns:
 - Use of comparative subtraction phrasing
 - Temporal sequence in verb tenses
- Final Response:

“The final sheep count is 6. Calculation steps: 1) Start with 17 sheep. 2) After delivery, ‘all but 9’ sold means 9 remain ($17-8=9$). 3) Later sold 3 more ($9-3=6$).”

E. FEEDBACK AGENT OUTPUT

Role: Feedback Specialist

Analysis:

- Quality Score: 0.90
- Gaps Identified:
 - Ambiguity in interpreting ‘all but X’
 - Lack of verification for initial assumptions
- Improvement Areas:
 - Clarify numerical assumptions
 - Explore alternative interpretations
 - Provide explicit step-by-step justification
- Decision: No Refinement Needed
- Feedback Comments:
 - Analysis correctly identifies core concepts
 - Unresolved ambiguity weakens mathematical rigor
 - Recommend sensitivity analysis
 - Need explicit statement of problem parameters